

Agentic AI

Complete Roadmap — 0 → Industry Ready

22 Weeks • 15 Projects • Senior Engineer Reviewed • All Missing Pieces Added + New Sections

15 Projects **22 Weeks** **10–25 LPA Target** **1 Niche to Own**

What was fixed vs original PDF

- ✓ 17 weeks → 22 weeks (realistic timeline)
- ✓ 3 original phases added: Testing, Security, System Design
- ✓ 3 NEW phases added: MCP Deep Dive, Cost Optimisation, Interview Prep
- ✓ Tools patched: LangGraph, Agno/Phidata, OpenTelemetry, pgvector
- ✓ Every phase now has proper error handling + edge cases
- ✓ Focus: Projects > Theory • GitHub > Resume • Niche > Generic

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ORIGINAL	Content from original PDF	PATCHED	Updated / corrected	ADDED	Was completely missing	NEW	Net new — not in original
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Phase 01 — Python + Backend Foundation

Weeks 1–3

TOPICS

- Variables, loops, functions
- OOP + clean code patterns
- async/await + event loop
- File I/O + JSON
- REST APIs with httpx / requests
- pytest — unit testing
- pydantic — data validation
- .env + secret management

PROJECTS

P0
1

Notes API (CRUD + MySQL)

Full CRUD with FastAPI + MySQL. Input validation with pydantic, proper HTTP error codes, pytest tests covering happy + error paths.

[\[FastAPI\]](#) [\[MySQL\]](#) [\[pydantic\]](#) [\[pytest\]](#)

P0
2

User Auth System

JWT login/signup with refresh tokens, rate limiting, bcrypt password hashing, and Redis session blacklist. No shortcuts.

[\[JWT\]](#) [\[bcrypt\]](#) [\[Redis\]](#) [\[rate-limiting\]](#)

Senior Tip

Do NOT touch LangChain until you can call the OpenAI API raw. Build async httpx calls from scratch — understand what wrappers are hiding.

Phase 02 — LLM APIs + Prompt Engineering

Week 4

TOPICS

- OpenAI API — raw calls, no wrappers first
- Prompt engineering: CoT, few-shot, system prompts
- Streaming responses with SSE
- Rate limit + retry logic with exponential backoff
- Token counting + cost tracking
- Structured output / JSON mode

PROJECTS

P0
3

AI Chatbot (memory + streaming)

Raw OpenAI calls with streaming, conversation memory, retry on rate limits, token budget tracking, and structured JSON output for tool-like responses.

[\[OpenAI\]](#) [\[streaming\]](#) [\[asyncio\]](#) [\[cost-tracking\]](#)

Warning

Missing from original PDF: no mention of retry logic or cost control. In production, unhandled rate limits will crash your agent at 2am.

Phase 03 — RAG System

Weeks 5–7

TOPICS

- Embeddings — OpenAI / HuggingFace
- Chunking: fixed-size + semantic
- Hybrid search: dense + BM25 sparse
- Evaluation: RAGAS metrics (faithfulness, context recall)
- Vector DB: Pinecone / Qdrant / pgvector
- Retriever + Generator pipeline
- Metadata filtering
- PostgreSQL + pgvector as lightweight alternative

PROJECTS

P0
4

PDF Chatbot with Citations

RAG pipeline with source citations, hybrid search, metadata filtering. Evaluate with RAGAS before and after tuning. Benchmark retrieval quality. Optionally replace Pinecone with pgvector to learn cost-effective setups.

[\[Qdrant\]](#) [\[pgvector\]](#) [\[LlamaIndex\]](#) [\[RAGAS\]](#) [\[HuggingFace\]](#)

Senior Tip

Most juniors build RAG and never measure if it works. RAGAS scores are the difference between a demo and a product.

Patched

pgvector added — PostgreSQL + pgvector is production-grade and free. You don't need Pinecone for most use cases.

Phase 04 — Advanced RAG

Weeks 8–9

TOPICS

- Reranking with Cohere
- Query expansion + HyDE
- Multimodal RAG (images + text)
- Contextual compression

• Guardrails + output validation

• End-to-end streaming from RAG pipeline

PROJECTS

P0
5

Multi-file AI Assistant

Multi-file RAG with reranking, query expansion, contextual compression, and output guardrails. Show benchmark numbers for each technique added. Add streaming from retrieval to response.

[\[Cohere\]](#) [\[guardrails\]](#) [\[HyDE\]](#) [\[evaluation\]](#) [\[streaming\]](#)

Senior Tip

Add streaming at the RAG level here — not just at the API level. This is a gap most tutorials miss.

Phase 05 — AI Agents **[PATCHED]**

Weeks 10–11

TOPICS

- LangChain Agents — basics
- Tools: search, code execution, APIs
- MCP protocol — intro
- Agent observability with LangSmith
- End-to-end streaming agent responses
- LangGraph — industry standard for agent workflows (2026)
- Memory: short-term (buffer) + long-term (vector)
- Prompt injection attack patterns + defenses
- Tool call failure handling + fallback chains
- Token streaming mid-execution (LangGraph)

PROJECTS

P0
6

Autonomous AI Agent (search + tasks)

Search + task agent built with LangGraph state machines. Add LangSmith tracing, graceful tool failure recovery, end-to-end streaming, and a prompt injection test suite.

[\[LangGraph\]](#) [\[LangSmith\]](#) [\[MCP\]](#) [\[prompt-injection\]](#) [\[streaming\]](#)

Patched

LangGraph replaces basic LangChain agents as the 2025/26 industry standard. Added security concerns — prompt injection is the #1 agentic AI risk.

NEW — Added

Streaming agent responses added — LangGraph streaming is a separate challenge from API-level streaming. Cover it [here](#).

Phase 06 — Multi-Agent Systems [PATCHED]

Weeks 12–13

TOPICS

- CrewAI / AutoGen — multi-agent basics
- Agent roles, handoff logic, task delegation
- Shared memory between agents
- A2A (Agent-to-Agent) protocol basics
- Agno (Phidata) — lighter, newer multi-agent framework
- Supervisor pattern for orchestrating agents
- Per-agent cost + latency monitoring
- Agent-to-agent communication design

PROJECTS

P0
7

Research Agent System (pick 1 niche)

Legal / Finance / HR / Medical document agent. Use supervisor pattern with LangGraph. Log per-agent cost and latency. Unit test agent outputs. Demonstrate A2A handoff between agents.

[Agno] [CrewAI] [supervisor] [niche-domain] [A2A]

Senior Tip

Pick ONE niche and go deep. A legal agent that actually works beats a generic agent every time in interviews and on your portfolio.

Patched

A2A protocol added. Companies like Google and Anthropic are pushing A2A as the standard — know it before your interview.

Phase 07 — Testing + Observability [ADDED]

Week 14

TOPICS

- pytest for agent pipelines
- LangSmith / Arize Phoenix tracing and spans
- CI-integrated eval pipelines
- Arize Phoenix — open source alternative to LangSmith
- Mocking LLM calls in tests (no real API calls)
- Golden test sets + regression datasets
- OpenTelemetry basics for AI apps
-

PROJECTS

P0
8

Evaluation Suite for P07 Agent

Write a full eval suite for your research agent. Mock all LLM calls, build a golden test set, track regressions across agent versions. Add OpenTelemetry spans. This single project separates junior devs from senior engineers.

[\[pytest\]](#) [\[mocking\]](#) [\[LangSmith\]](#) [\[Arize\]](#) [\[OpenTelemetry\]](#) [\[CI/CD\]](#)

Added — Was Missing

This entire phase was missing from the original PDF. No testing = no production. Every serious AI company runs eval suites before every deployment.

NEW — Added

Arize Phoenix added — it's open source, free, and increasingly preferred over LangSmith in production. Know both.

Phase 08 — Production + Deployment

Weeks 15–16

TOPICS

- FastAPI advanced: background tasks, dependencies
- Docker + docker-compose
- Background tasks: Celery / ARQ
- GitHub Actions CI/CD pipeline
- JWT auth + structured JSON logging
- Deploy: Render / Railway / AWS
- Redis for caching + job queues
- Health check + readiness endpoints

PROJECTS

P0
9

Deploy AI SaaS Backend

Full FastAPI backend with Redis queue, Celery workers, Docker, GitHub Actions CI, structured JSON logs, and health check endpoints. Must pass CI before merge.

[\[Docker\]](#) [\[GitHub Actions\]](#) [\[Redis\]](#) [\[Celery\]](#)

Senior Tip

Do not skip structured JSON logging. grep and CloudWatch logs are 10x easier to debug when every log line is valid JSON.

Phase 09 — Security for Agentic AI **[ADDED]**

Week 17

TOPICS

- Prompt injection: direct + indirect attacks
- Tool call sandboxing (no raw exec)
- API key rotation and secrets management
- OWASP LLM Top 10 — all 10 risks

- PII detection and redaction in RAG pipelines

- Output validation + content filtering

PROJECTS

P1
0

Security Hardening Sprint

Audit your P07 agent for prompt injection vulnerabilities. Add sandboxed tool execution, scan RAG pipeline for PII leaks, document a threat model. Run OWASP LLM checklist against your entire codebase.

[\[OWASP\]](#) [\[sandboxing\]](#) [\[PII-detection\]](#) [\[threat-model\]](#)

Added — Was Missing

This phase was completely missing from the original PDF. Prompt injection is the single biggest security risk in agentic AI — it cannot be an afterthought.

Phase 10 — MCP + Agent Protocols Deep Dive [NEW]

Week 18

TOPICS

- MCP (Model Context Protocol) — full spec
- Building a custom MCP server from scratch
- A2A (Agent-to-Agent) protocol — Google standard
- ACP (Agent Communication Protocol)
- Tool exposure via MCP — resources, prompts, tools
- MCP client integration with LangGraph agents
- Multi-agent orchestration via protocols
- Protocol versioning + backward compatibility

PROJECTS

P1
1A

Custom MCP Server

Build a fully spec-compliant MCP server that exposes at least 3 tools (e.g., database query, file read, web search). Connect it to your P06 LangGraph agent. Document the protocol handshake.

[\[MCP\]](#) [\[FastAPI\]](#) [\[LangGraph\]](#) [\[protocol-design\]](#)

NEW — Added

This entire phase is new — was not in the original PDF at all. MCP is now the standard way companies expose tools to agents. Interviewers at Anthropic, Google, and Amazon are already asking about it.

Senior Tip

Don't just use MCP — build a server. Understanding the server side is what separates you from devs who only follow tutorials.

Phase 11 — Cost Optimisation + Model Routing [NEW]

Week 19

TOPICS

- Model routing: cheap vs expensive LLM per task
- Prompt caching (Anthropic / OpenAI)
- Batching API calls to reduce cost
- Token budget management per agent
- Cost dashboard — track per-run spend
- LiteLLM — unified interface across providers
- Fallback chains: GPT-4 → GPT-3.5 on rate limit
- Latency vs cost tradeoffs in production

PROJECTS

P1
1B

Cost Dashboard + Model Router

Build a middleware layer that routes LLM calls by complexity — cheap model for simple tasks, expensive for complex. Add a real-time cost dashboard showing per-agent, per-run spend. Integrate prompt caching.

[\[LiteLLM\]](#) [\[Redis\]](#) [\[FastAPI\]](#) [\[cost-tracking\]](#) [\[model-routing\]](#)

NEW — Added

This phase was completely missing from the original. In production, uncontrolled LLM costs can bankrupt a startup. Every senior AI role will ask how you controlled costs.

Senior Tip

LiteLLM is the de facto standard for multi-provider routing. Learn it — it also makes your code provider-agnostic.

Phase 12 — Frontend Integration

Weeks 20–21

TOPICS

- Streamlit / Gradio — fast demo UIs (2 days max)
- React / Next.js — optional for backend/AI roles
- WebSockets for real-time agent updates
- Streaming UI (SSE + token-by-token display)
- Error states + loading indicators
- Agent execution status UI

PROJECTS

P1
2

Full AI SaaS App

Streamlit or Next.js frontend connected to your agent backend. WebSocket streaming, error states, loading indicators. If targeting pure backend roles, Streamlit in 2 days is enough — no bloat.

[\[Next.js\]](#) [\[Streamlit\]](#) [\[WebSockets\]](#) [\[streaming-UI\]](#)

Senior Tip

If your goal is AI/Backend roles, don't spend 2 full weeks on Next.js. Use Streamlit as a demo wrapper and focus time on agents and protocols.

Phase 13 — System Design for AI [ADDED]

Week 22

TOPICS

- RAG vs fine-tuning — when to use which
- Multi-tenant AI architecture patterns
- Async vs sync agent execution patterns
- Agent state persistence patterns
- Scaling vector DBs: sharding + replication
- Cost optimisation: caching, batching, model routing
- MCP server at scale — load balancing agents
- Failure recovery in multi-agent pipelines

PROJECTS

Added — Was Missing

No project here — this is interview prep. Study these topics and practice explaining them. Every senior AI role will ask system design questions. White-board at least 3 of these with a friend.

Senior Tip

The most common system design question in 2026 AI interviews: 'Design a multi-tenant RAG system that handles 10k users without data leakage.' Practice it.

Phase 14 — Interview Prep + Career Launch [NEW]

Week 23

TOPICS

- LinkedIn profile optimisation for AI roles
- GitHub portfolio presentation strategy
- LangChain / LangGraph common interview questions
- Salary negotiation for AI developer roles
- Cold outreach message templates (HR + CTO)
- System design interview practice (Pramp, Exponent)
- Behavioural questions for AI roles
- Targeting product companies vs service companies

PROJECTS

P1
3

Career Launch Package

Produce: (1) optimised LinkedIn profile, (2) GitHub README for each project with CI badges + RAGAS scores, (3) 3 cold outreach templates, (4) 1 system design write-up for your niche domain. This is not optional — no portfolio = no callbacks.

[LinkedIn] [GitHub] [portfolio] [outreach]

NEW — Added

This phase was completely absent from the original PDF. Building 12 projects means nothing if you cannot present them. Recruiters spend 20 seconds on a GitHub profile — make it count.

Senior Tip

Template for cold outreach that works: 'Hi, I built [specific agent] that does [specific thing]. I noticed you're hiring for [role]. Here is the GitHub link.' One line. One link. No fluff.

Phase 15 — Final Capstone **[PATCHED]**

Week 24

TOPICS

- End-to-end agentic product in 1 niche domain
- Security hardened (OWASP checklist passed)
- Cost dashboard with per-run tracking
- Full test suite + RAGAS eval + CI pipeline
- MCP server exposing agent tools
- Public GitHub + demo video + write-up

PROJECTS

P1
5

Production Agentic AI Product

Your niche agent (Legal / Finance / HR / Medical) deployed end-to-end. Full test suite, CI/CD, structured logs, security hardened, RAGAS scores documented, MCP server integrated, cost dashboard live, and a 3-minute demo video. This is your portfolio centrepiece.

[full-stack] [agentic] [production] [portfolio] [MCP] [cost-dashboard]

Senior Tip

1 great niche project beats 10 generic ones. GitHub > Resume. Projects > Certificates. Demo video is non-negotiable — it's your async interview.

Patched

MCP integration and cost dashboard added to capstone requirements. These are now table stakes in senior AI interviews.

Industry Tools — Must Know

Category	Tools	Status
Language + API	Python, FastAPI, httpx, pydantic	Original
LLM APIs	OpenAI, Anthropic, HuggingFace, LiteLLM	Patched (LiteLLM added)
Vector DBs	Pinecone, Qdrant, ChromaDB, pgvector	Patched (pgvector added)
RAG Frameworks	LlamaIndex, LangChain	Original
Agent Orchestration	LangGraph, LangChain Agents	Patched (LangGraph added)
Multi-Agent	CrewAI, AutoGen, Agno/Phidata	Patched (Agno added)
Agent Protocols	MCP, A2A, ACP	NEW — was missing
Testing	pytest, RAGAS, LangSmith, Arize Phoenix	Added — was missing
Observability	OpenTelemetry, LangSmith, Arize Phoenix	NEW — was missing
Cost Management	LiteLLM, prompt caching, model routing, cost dashboards	NEW — was missing
Security	OWASP LLM Top 10, Guardrails AI	Added — was missing
Infrastructure	Docker, GitHub Actions, Redis, Celery	Partially added
Cloud / Deploy	AWS/GCP basics, Render, Railway	Original
Frontend	Streamlit, Gradio, React/Next.js	Streamlit added

Final Outcomes + Career Path

15 Projects	Built across all 15 phases — from Notes API to Production Agentic AI.
24 Weeks	Realistic timeline. Not the 17-week marketing claim in the original.
10–25 LPA	AI Engineer / Backend roles in the Indian market. Product companies > service companies.
1 Niche Domain	Legal / Finance / HR / Medical — deep beats wide, always.
Public GitHub	All 15 projects live with README, CI badges, RAGAS scores, and tests.
MCP Server	Custom MCP server live — separates you from 95% of candidates.

Cost	
Dashboard	Per-run cost tracking on capstone — shows you think like a senior.
Eval Suite	RAGAS scores + pytest suite + LangSmith traces documented.
Security	
Hardened	OWASP LLM checklist passed on capstone project.
Demo Video	3-minute walkthrough of P15 capstone — your async interview asset.
Career Package	LinkedIn profile, cold outreach templates, system design write-up — ready to apply.

[Projects](#) > [Theory](#) • [GitHub](#) > [Resume](#) • [Niche](#) > [Generic](#) • [Build](#) > [Watch](#)